Problem 1

Task: Obtain the closed-loop transfer function $C(s)/R(s)$ by simplifying block diagram of the SISO system shown in Fig. 1 .

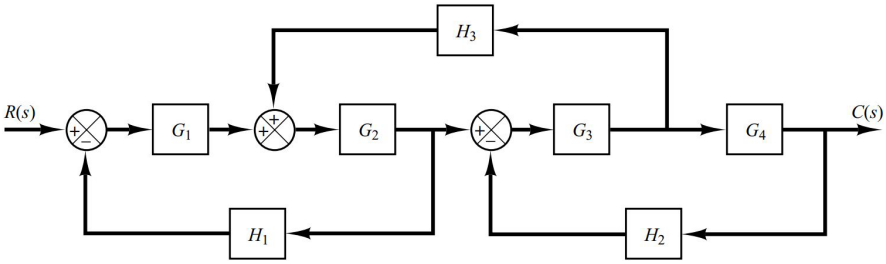


Figure 1: P1: Block Diagram

Solution:

1. Move branch between G_3 and G_4 to the right of G_4. Move summing junction between G_1 and G_2 to the left of G_1 as shown in Fig. 2

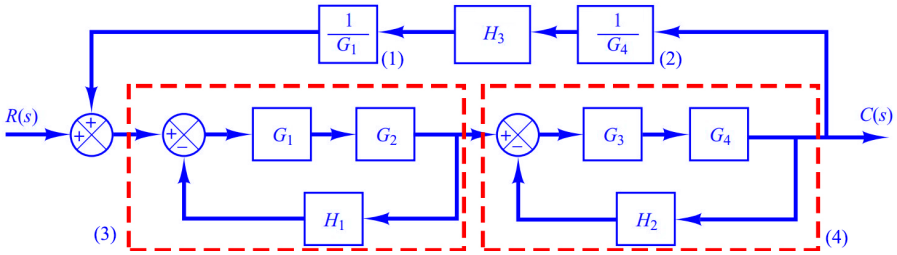


Figure 2: Step 1

2. Simplify the negative feed back loops (3 and 4) marked in red rectangles (Fig. 2) using the feedback rules given in Figure 3. The result is shown in Fig. 4

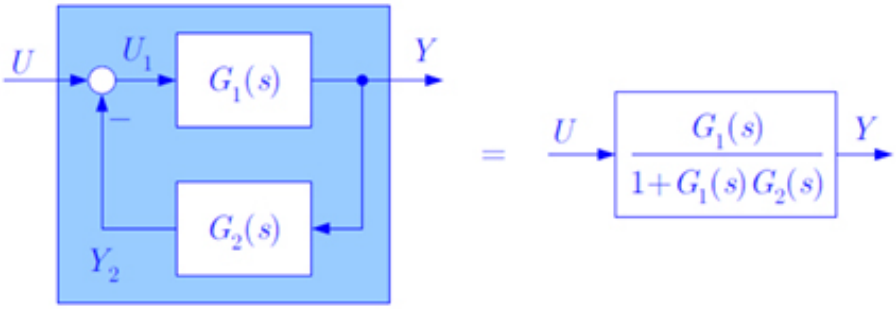


Figure 3: Simplification of Negative Feedback Loop

3. The same negative feedback rule (Fig. 3) can be applied to block diagram in Fig. 4 to achieve the final result in Fig. 5.

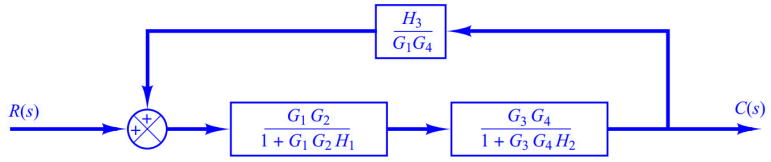


Figure 4: Simplification of Feedback Loops

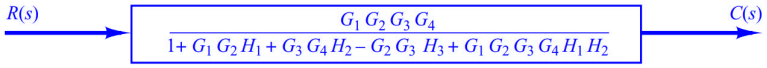


Figure 5: Solution

Problem 2:

Task: Obtain the transfer function $Y(s)/U(s)$ for the mechanical system shown in Fig. 6 with u as a displacement input.

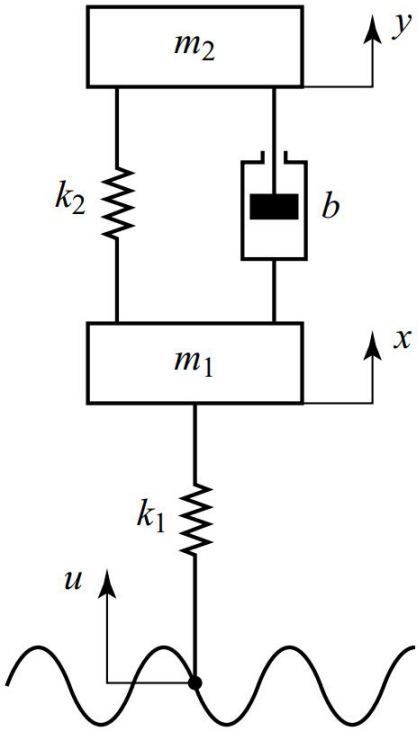


Figure 6: P2

1. **Equation of Motion**: Apply Newton's second law to get the equation of motion for each mass. Assume that x and y are measured from steady-state positions attained under the influence of gravity.

$$\begin{aligned} m_1\ddot{x} &= k_2(y-x) + b(\dot{y}-\dot{x}) + k_1(u-x) \\ m_2\ddot{y} &= -k_2(y-x) - b(\dot{y}-\dot{x}) \end{aligned} \tag{1}$$

Reordering x and y

$$\begin{aligned} m_1\ddot{x} + b\dot{x} + (k_1 + k_2)x = k_2y + b\dot{y} + k_1u \\ m_2\ddot{y} + b(\dot{y} + k_2y = k_2x + b\dot{x} \end{aligned} \tag{2}$$

2. **Transfer Function:** Apply Laplace transform to the equations in(2), assuming zero initial conditions.

$$[m_1s^2 + bs + (k_1 + k_2)]X(s) = (bs + k_2)Y(s) + k_1U(s) \tag{3}$$

$$[m_2s^2 + bs + k_2]Y(s) = (bs + k_2)X(s) \tag{4}$$

from (4) $X(s)$ can be written as

$$X(s) = \frac{m_2s^2 + bs + k_2}{bs + k_2}Y(s) \tag{5}$$

From (5) and (3)

$$[m_1s^2 + bs + (k_1 + k_2)]\frac{m_2s^2 + bs + k_2}{bs + k_2}Y(s) = (bs + k_2)Y(s) + k_1U(s) \tag{6}$$

simplifying (6) yields

$$\frac{Y(s)}{U(s)} = \frac{(bs + k_2)k_1}{m_1m_2s^4 + (m_1 + m_2)bs^3 + (k_1m_2 + m_1k_2 + m_2k_2)s^2 + k_1bs + k_1k_2} \tag{7}$$

3. **Simulation:**
see Matlab file Assignment_P2.m

Problem 3:

The equations of motions are

$$\begin{aligned} J_l\ddot{\theta}_l + B_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0 \\ J_m\ddot{\theta_m} + B_m\dot{\theta}_m - k(\theta_l - \theta_m) = u \end{aligned} \tag{8}$$

Reorganizing (8

$$\begin{aligned} \ddot{\theta}_l = -\frac{B_l}{J_l}\dot{\theta}_l - \frac{k}{J_l}(\theta_l - \theta_m) \\ \ddot{\theta_m} = -\frac{B_m}{J_m}\dot{\theta}_m - \frac{k}{J_m}(\theta_l - \theta_m) + \frac{1}{J_m}u \end{aligned} \tag{9}$$

State-space model with state x and output y is given by

$$\begin{aligned} \dot{x} = Ax + Bu \\ y = Cx + Du \end{aligned} \tag{10}$$

Considering the state vector $x = [\theta_l \quad \dot{\theta}_l \quad \theta_m \quad \dot{\theta}_m]$, equation (9) can be written as

$$\begin{bmatrix} \dot{\theta}_l \\ \ddot{\theta}_l \\ \dot{\theta}_m \\ \ddot{\theta}_m \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{J_l} & -\frac{B_l}{J_l} & \frac{k}{J_l} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k}{J_l} & 0 & -\frac{k}{J_m} & -\frac{B_m}{J_m} \end{bmatrix}}_{A} \begin{bmatrix} \theta_l \\ \dot{\theta}_l \\ \theta_m \\ \dot{\theta}_m \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_m} \end{bmatrix}}_{B} u$$
$$y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}}_{C} \begin{bmatrix} \theta_l \\ \dot{\theta}_l \\ \theta_m \\ \dot{\theta}_m \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{D} u \tag{11}$$

Problem 4:

1. The general form of first order transfer function can be expressed as

$$G(s) = \frac{K}{\tau s + 1} \tag{12}$$

where K is teh gain and τ is time constant. Comparing $G(s) = \frac{80}{s+40}$ with (12) gives $K = 80/40 = 2$ and $\tau = 1/40 = 0.025$. The first order transfer function of the system in problem 4 then can be written as

$$G(s) = \frac{2}{0.025s + 1} \tag{13}$$

2. A general form of step response $c(t)$ of a first order system is given as

$$c(t) = K(1 - e^{-t/\tau}) \tag{14}$$

Using K and τ from the previous step gives

$$c(t) = 2(1 - e^{-t/0.025}) = 2 - 2e^{-40t} \tag{15}$$

3. see Matlab code Assignmnet_P4.m

Problem 5:

1. **Transfer Function:** Equation of motion for mass spring damper system is given by

$$m\ddot{x} + b\dot{x} + kx = F \tag{16}$$

Laplace transform (with zero initial conditions) yields

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \tag{17}$$

Transfer Function is given by

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} \tag{18}$$

2. General form of second order system transfer function can be written as

$$G(s) = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} + 1} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} \tag{19}$$

where K is gain, ω_n is natural frequency and ζ is damping. ω_n can be calculated as

$$\omega_n = \sqrt{\frac{k}{m}} \tag{20}$$

Using system parameters the transfer function in (18) can be written as

$$\frac{X(s)}{F(s)} = \frac{0.5}{s^2 + 4s + 50} \tag{21}$$

Comparing (21) and (18) ω_n and ζ can be calculated as

$$\begin{aligned} \omega_n^2 = 50 &\implies \omega_n \approx 7.0711 \\ 2\zeta\omega_n = 4 &\implies \zeta = \frac{4}{2\sqrt{50}} \approx 0.2828 \end{aligned} \tag{22}$$

Since $0 < \zeta < 1$, it can be concluded that the system has under-damped with two complex conjugate poles.

3. The general form of step response for second order system (with complex conjugate system poles $-\sigma_d \pm j\omega_d$) is given by

$$c(t) = K_0 + K_1 e^{-\sigma_d t} cos(\omega_d t + \phi) \tag{23}$$

Gain K_0 is calculated (using partial fraction)

$$K_0 = \frac{0.5}{50} = 0.01 \tag{24}$$

The system has two complex poles (using poles or roots command in Matlab)

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} = -2 \pm 6.7823k \tag{25}$$

The under-damped response is then given by (using 23)

$$c(t) = 0.01 - 0.01e^{-2t} cos(6.7832t + \phi) \tag{26}$$

4. Peak time T_p, rise time T_r , settling time T_s and percent overshoot $\%OS$ can be calculated using the following expressions

$$\begin{aligned} T_p &= \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \approx 0.4632 \\ T_s &= \frac{4}{\zeta\omega_n} \approx 2 \\ \%OS &= e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \approx 39.6\% \end{aligned} \tag{27}$$

Rise time can be calculated using normalized rise time Table(Figure 4.16 in the book). Interpolating the table data to find the normalized rise time NT_r at $\zeta = 0.2828$

$$NT_r = (\zeta - 0.2) * \frac{1.321 - 1.203}{0.1} + 1.203 = 1.3007 \tag{28}$$

the rise time T_r is then

$$T_r = \frac{NT_r}{\omega_n} = 0.184. \tag{29}$$

The same can be confirmed from the step response using Matlab. See sample code in Matlab File Assignment1_P5.m